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GIS-based stochastic modeling of physical accessibility using GPS-based floating car data and Monte Carlo simulation

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The term physical accessibility has long been used by geographers, economists, and urban planners and reflects the relative ease of access to/from several urban/rural services by considering the traveling costs. Numerous accessibility measures, ranging from simple to sophisticated, can be observed in the geographical information systems (GIS)-based accessibility modeling literature. However, these measures are generally calculated from a constant catchment boundary (a most likely or average catchment boundary) based on constant deterministic transportation costs. This is one of the fundamental shortcomings of the current GIS-based accessibility modeling and creates uncertainty about the accuracy and reliability of the accessibility measures, especially when highly variable speeds in road segments are considered. The development of a new stochastic approach by using global positioning system (GPS)-based floating car data and Monte Carlo simulation (MCS) technique could enable handling the variations in transportation costs in a probabilistic manner and help to consider all possible catchment boundaries, instead of one average catchment boundary, in accessibility modeling process. Therefore, this article proposes a stochastic methodology for GIS-based accessibility modeling by using GPS-based floating car data and MCS technique. The proposed methodology is illustrated with a case study on medical emergency service accessibility in Eskisehir, Turkey. Moreover, deterministic and stochastic accessibility models are compared to demonstrate the differences between the models. The proposed model could provide better decision support for the decision-makers who are supposed to deal with accessibility, location/allocation, and service/catchment area related issues.

Keywords: stochastic/probabilistic accessibility modeling; geographical information systems (GIS); global positioning systems (GPS); floating car data (probe vehicle data); Monte Carlo simulation

1. Introduction

The term physical accessibility has long been used by geographers, economists, and urban planners and basically reflects the relative ease of access to/from several urban/rural services by considering several traveling costs (Halden *et al.* 2000, Makri 2002, McGrail and Humphreys 2009). Physical accessibility measures are concerned with equity and a better distribution of people and activities in the territory and can be accepted as key variables for supporting supply/demand, location/allocation, and service/catchment area related planning

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policies and strategies at national, regional, and local levels (see Kuntay 1990, Juliao 1999, Halden *et al.* 2000, Radke and Mu 2000, Makri 2002).

As physical accessibility measures need organization of huge and complex spatial data sets, they often lend themselves to geographical information systems (GIS) for modeling. GIS have unique capabilities to handle spatially referenced information in a way which aids decision-making and provides a powerful interface for 'data storage, management, and manipulation capabilities for both spatial and attribute data,' 'core data analyses capabilities such as buffer, overlay, proximity, shortest path, raster cost-distance, etc.,' 'programming capabilities to handle current models or create new models,' and 'mapping and visualization capabilities to evaluate the results of the analyses' (Black *et al.* 2004).

Numerous accessibility measures, ranging from simple to sophisticated, can be found in the GIS-based accessibility modeling literature. The simple measures only consider proximity in terms of time and distance. The sophisticated ones consider both proximity and availability including the size of the supply and demand. Some of the most widely used accessibility measures in the literature are

- 'Travel time/distance measures, service/catchment areas' (travel time or distance to nearest supply/demand calculated from Euclidian/network-based costs),
- 'Cumulative opportunity measures' (consider the total amount of demand/supply inside the catchment areas),
- 'Population to provider ratio measures' (supply to demand ratios, calculated inside the catchment areas),
- 'Kernel density measures' (uses the Gaussian kernel approach to calculate the density value of each demand/supply),
- 'Gravity-based measures' (a combined indicator of accessibility and availability by considering the attractiveness of the supply/demand),
- 'Two-step floating catchment area (FCA) measures' (repeat the process of catchment area calculation twice for both supply and demand points), etc.

(For a detailed review, see Luo and Wang 2003, Guagliardo 2004, Bagheri *et al.* 2006, McGrail and Humphreys 2009.)

Whether simple or sophisticated, one of the fundamental shortcomings of the current GIS-based accessibility measures is that they are generally calculated from a constant catchment boundary (average or maximum catchment boundary) based on deterministic traveling costs such as 'Euclidian distance costs' or 'constant transportation network-based costs' (Figure 1). The Euclidian distance costs are simplified unconstrained costs and generally based on bird flight distances such as buffer polygons and Voronoi/Thiessen polygons. Constant transportation network-based costs are generally constrained by the road network and calculated from average or maximum speeds on classified road segments such as 120 km/h for highways, 50 km/h for main streets, and 30 km/h for local streets (see several examples, Ritsema van Eck and de Jong 1999, Makri 2002, Luo and Wang 2003, Bixby 2004, Lotfi and Koohsari 2009, McGrail and Humphreys 2009, Vahidnia *et al.* 2009).

Although the Euclidian distance costs and constant (fixed) average travel speed costs are widely used in GIS-based accessibility modeling literature, such deterministic costs are not realistic, especially when considering the highly variable speeds in road segments and having uncertainty about the accuracy and reliability of the accessibility measures in terms of deterministic catchment boundaries. Therefore, *the aim of this article* is to develop a stochastic methodology for GIS-based accessibility modeling process that could handle

Euclidian distance cost-based service/catchment areas

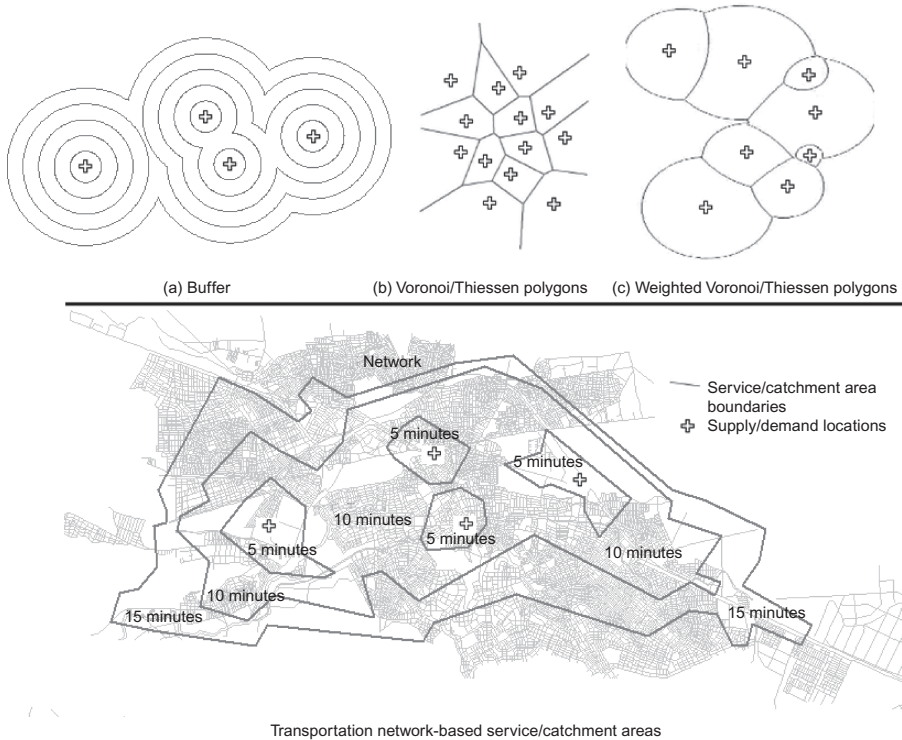


Figure 1. Constant catchment boundaries based on deterministic traveling costs.

variations in traveling costs and consider all possible catchment boundaries, instead of one average or maximum catchment boundary.

The main *benefit of the proposed stochastic methodology* is that the proposed stochastic model could provide additional information related to the accuracy and the reliability of the catchment boundaries in accessibility modeling, which means better decision support for the decision-makers who are supposed to deal with accessibility, location/allocation, and service/catchment area related issues. The proposed stochastic model allows systematic treatment of uncertainties related to the catchment boundaries and the crisp catchment boundaries in the deterministic model turn into probabilistic catchment boundaries providing decision-makers to operate different levels of uncertainty in modeling of accessibility.

The proposed stochastic methodology is illustrated with a case study on medical emergency service accessibility, in Eskisehir, Turkey, and the results of the deterministic and stochastic accessibility models are compared. The main focus of the case study is not to evaluate a specific accessibility condition in a detailed manner but to provide a methodological discussion and comparison between the deterministic and stochastic accessibility modeling process.

2. Proposed stochastic methodology

The proposed stochastic accessibility model consists of four major steps: (1) data collection, (2) generation of speed surface, (3) extraction of speed statistics, and (4) Monte Carlo simulation (MCS) (Figure 2).

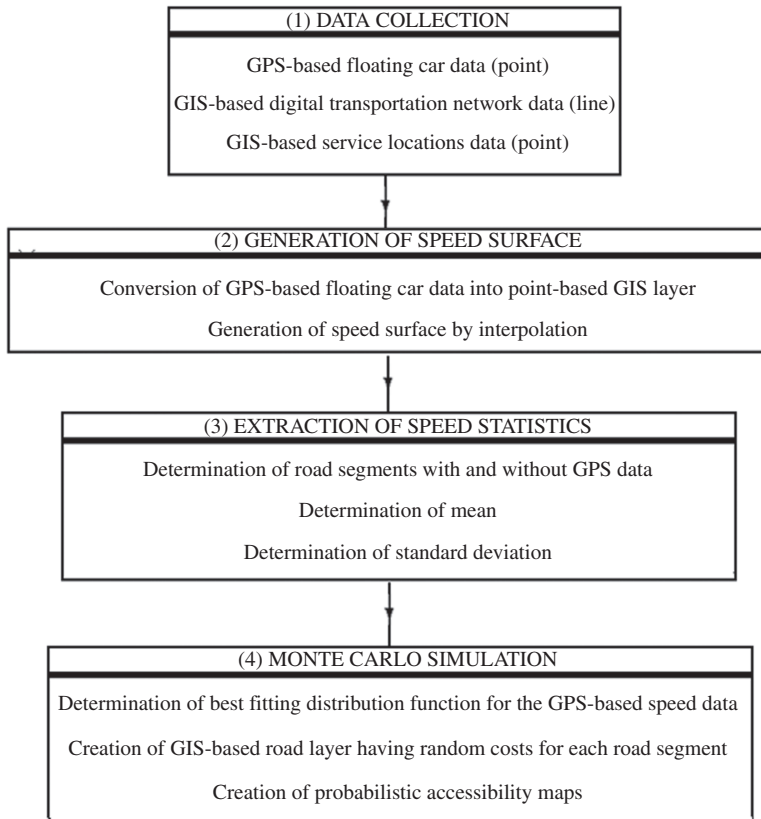


Figure 2. The flowchart of the methodology.

2.1. Data collection

There are three basic data used in the model:

- Global positioning system (GPS)-based floating car data, which are in the form of point objects (includes location, speed, and time information with a predefined time or distance interval),
- GIS-based digital transportation network data, which are in the form of line objects (includes the road lengths/widths and location of road segments/junctions and basic classification of roads according to their types such as highways, boulevards, and main street),
- GIS-based service locations data, which are in the form of point objects (includes the location of the service locations).

The GPS-based floating car data are used in the model to understand the speed variations on road segments in terms of mean and standard deviation. Although there are several traffic data collection methods such as stationary traffic sensors (induction loops, optical systems), space and airborne techniques (observation from planes, satellites), and GPS-based floating car data (GPS probe vehicle data), the GPS-based floating car data method is one of the most suitable traffic data collection methods. The GPS-based floating car data method is based on

recording position and speed from vehicle(s) moving in the traffic. The GPS-based floating car data, when integrated with GIS, can provide variations in transportation costs. Compared with conventional traffic data collection techniques, the GPS technique is able to provide speed information in a continuous manner with several detail and complexity depending on the track intervals (e.g., every 10 seconds or every 50 meters), day and time preferences (e.g., annual, seasonal, time-based variations: rush hours, normal hours, weekdays, or weekends, etc.), methodology (e.g., data collection from whole transportation network or only the main road segments), and accuracy (e.g., ranging from meters to centimeters based on the used GPS instrument). Moreover, such data collection is relatively fast and cheap as well as providing accurate position and speed with availability to be integrated in GIS (see several examples and detailed explanation in Zito *et al.* 1995, D'Este *et al.* 1999, Quiroga 2000, Taylor *et al.* 2000, Derekenaris *et al.* 2001, Mintsis *et al.* 2004, Daoqin *et al.* 2009).

The GPS-based floating car data can generally be obtained from the transportation department of municipalities. The detail and complexity of the GPS data and the considered traffic conditions in the model (seasonal variations, rush hour variations, etc.) are mostly based on the aim, the budget, and the detail needs of the study. For example, a research which requires accurate determination of catchment boundaries in rush hour time could need GPS data collected in long term (1 year or more) from the entire road segments for rush hour time with frequent track intervals or could support the GPS data with the data from other traffic data collection methods. However, GPS data collected only from the main street segments with large track intervals without considering a specific time or seasonal period could be enough for another research which does not require high accuracy and reliability in determination of catchment boundaries.

Although the proposed stochastic model is performed by using GPS-based floating car data, data from the other traffic data collection methods such as induction loops and optical systems can also be integrated into the stochastic model and provide complementary traffic profile information, where it is not possible or partly possible to obtain GPS-based floating car data.

The GIS-based digital transportation network and GIS-based service locations are the other basic data needed for modeling of accessibility and can be obtained from the related department of municipalities (e.g., department of transportation, planning).

2.2. Generation of speed surface

Generation of a continuous speed surface is used in the model to extract the speed variation (mean and standard deviation statistics of each road segment) from the GPS-based floating car data and integrate with the transportation network. The GPS-based floating car data are generally obtained in log file format and need to be converted to a point data having attributes of x and y coordinates and speeds at that point. The continuous speed surface is created from the GPS-based floating car data by using the spatial interpolation and the buffer capabilities of GIS (Figure 3).

The inverse distance weighted (IDW) interpolation technique is used in the generation of speed surface. The aim of the interpolation is to fill the speed gaps between the known GPS points and create a continuous speed surface for the road segments that have GPS data. The IDW technique is mainly based on the assumption that the unknown cells are more alike to closer cells than those are farther. During the interpolation process, the road segments in the transportation network that have GPS data are buffered as their road widths and used as a boundary/mask object in the interpolation process. Using a boundary/mask object for the road segments that have no GPS data is a vital step in the interpolation process, as the speed surface is created for only the street segments that have GPS-based floating car data.

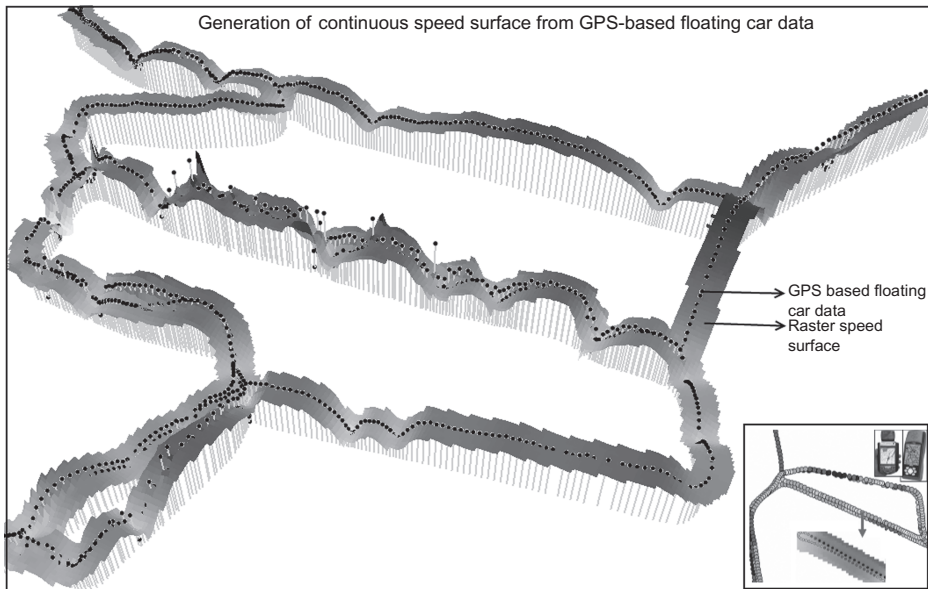


Figure 3. Production of raster speed surface from GPS data by using interpolation.

2.3. Extraction of speed statistics

Extraction of speed statistics step covers calculation of the mean and standard deviation statistics of speed data from the continuous speed surface for each of the road segments in the transportation network by using 'buffer,' 'join,' and 'zonal statistics' capabilities of GIS. For each of the road segments that have GPS data, the mean and standard deviation statistics of the speed are extracted from the speed surface. For each of the road segments that do not have GPS data, the average statistics of the GPS data according to their road types (highway, boulevard, street, etc.) are assigned.

2.4. Monte Carlo simulation

Stochastic transportation cost calculation approaches can be incorporated into the accessibility analyses by using simulation. The word 'simulation' refers to analysis of the effect of varying inputs on outputs of the modeled system. A simulation involves hundreds or thousands of evaluations of the model for all possible inputs and gives a probabilistic measure of the outputs. Monte Carlo simulation (MCS) method is a well-known method to create the random realizations of a deterministic model (Metropolis and Ulam 1949, Hoffman 1998). By integrating MCS method into GIS-based accessibility modeling process, possible random transportation cost values can be used instead of constant deterministic costs. Hence, the probability of an accessibility outcome can be obtained in terms of all possible catchment boundaries. By this way, accessibility can be expressed in terms of probability of having a certain accessibility measure instead of stating a deterministic accessibility measure. The probabilistic accessibility measures can take the uncertainties of transportation costs into account and enhance decision-making processes due to consideration of variability involved in the transportation cost parameters.

The aim of the MCS in the stochastic model is to produce random costs for the accessibility model. To perform MCS, there is a need to determine the best fitting probability density function for the speed data. Therefore, GPS-based floating car data are statistically analyzed by using Easy Fit distribution fitting software. Several probability density functions are tested and ranked by using the goodness of fit tests of Kolmogorov–Smirnov, Anderson–Darling, and chi-squared. The results showed that the best fitting probability density functions for the speed data are generalized extreme value, normal, Weibull, Johnson, beta, log-logistic, and log-normal probability density functions. As the results are significantly close to the normal distribution, the normal distribution is used as input distribution function to produce random costs in MCS.

‘The best fitting distribution function,’ ‘the extracted speed statistics of mean and standard deviation,’ and ‘the length of the road segments’ are used as input variables in MATLAB programming environment and ‘random time costs for each road segment’ is produced as an output (Table 1).

As a final step, accessibility is modeled for each of the MCS-based random costs by using ‘service area function’ and ‘model builder’ capabilities of GIS. The service area function connects points of equal travel time away from a service or services on a transportation network and creates polygons which represent the catchment area boundaries. If a service is defined as the reference point, polygons can be drawn connecting points in all directions that can be reached within a threshold time or distance. Locations inside the polygons are determined as accessible and outside the polygons are determined as inaccessible. The proposed stochastic model calculates these catchment area boundaries for each of the MCS-based random costs for a defined time threshold (5 minutes in the case study) and converts each of the calculated boundary polygons to a binary raster map with a classification of ‘1’ for accessible and ‘0’ for inaccessible areas in model builder environment.

The sum of these binary raster maps are used to produce a final stochastic accessibility map, which has probability scores of accessibility in terms of all possible combination of catchment boundaries. For example, if 1000 simulations are performed to model accessibility of emergency services for a defined threshold of 5 minutes, the pixel value of ‘0’ means there is 0% probability, 500 means there is 50% probability, and 1000 means there is 100% probability that a particular cell have 5 minutes emergency service accessibility.

3. Case study

The proposed methodology is implemented in a case study on medical emergency service accessibility in Eskisehir, which is one of the biggest cities of Turkey with an urban population of nearly 620,000 according to official population census of 2009 (Figure 4).

Medical emergency service accessibility reflects the level of medical emergency services to reach their catchment areas within critical time thresholds and helps to identify the critical areas that are out of service range. The ‘5 minutes’ is determined as threshold cost in modeling of medical emergency service accessibility as it is a critical time for saving lives from medical emergency response point of view (Blackwell *et al.* 2009).

Although the case study is implemented on medical emergency service accessibility, the primary focus of the study is not to evaluate a specific accessibility condition in a detailed manner but to provide a discussion and comparison between the deterministic and stochastic accessibility modeling in terms of accuracy and reliability of the catchment areas.

Table 1. Random costs of time in 'seconds' for each road segment.

Road segment_ID	Length	Speed_mean	Speed_standard deviation	Random cost 1	Random cost 2	Random cost 3	Random cost 4	Random cost 5	Random cost 6	Random cost 7	Random cost 8	Random cost 9	Random cost 10
0	293.5085	30.3851	10.859	41.24893	86.15652	33.37727	31.6213	59.07801	24.46133	24.47207	35.34768	31.22049	32.82367
1	250.6248	30.3851	10.859	22.65811	22.34145	17.04076	27.0898	25.65112	44.74891	60.25391	32.63668	22.202	30.28908
2	548.5082	54.6999	12.4341	51.77828	42.6084	37.44436	34.26819	36.84247	25.94823	26.44899	31.68833	35.4606	44.34082
3	1066.366	30.3851	10.859	98.19699	121.924	64.11029	110.4679	237.7253	111.4147	107.5058	154.9892	116.8837	161.1202
4	770.5008	30.3851	10.859	86.12195	77.01633	114.1006	82.72607	162.9962	98.20128	102.6084	68.21524	113.1037	135.0271
5	501.0478	54.6999	12.4341	59.80753	37.75437	57.93948	33.90026	47.18519	39.03712	41.60176	34.27442	27.20955	24.4512
6	376.6355	30.3851	10.859	55.35959	37.42892	29.18221	53.44374	54.39972	31.04412	44.10367	35.3051	31.07877	40.35177
7	44.12451	54.6999	12.4341	4.043421	1.819565	3.665775	4.033911	2.724377	3.491626	3.935074	4.290394	3.364814	2.604899
8	76.21749	54.6999	12.4341	5.17032	5.032397	5.578908	5.263328	4.109721	4.546001	7.29297	4.735616	3.140697	5.092461
9	36.01389	30.3851	10.859	8.115594	6.980663	4.050842	8.990352	3.448721	3.788088	3.516604	4.16835	3.086868	5.606012
10	41.98066	30.3851	10.859	3.758177	6.258278	4.119564	4.666138	19.10712	4.867858	5.215647	4.291541	4.829063	19.81016



Figure 4. The case study area.

3.1. Data

The following data were collected to implement the proposed stochastic model:

- Digital transportation network data of Eskişehir Metropolitan area and their hierarchies
- The location of medical emergency service stations in Eskişehir
- GPS-based floating car data of Eskişehir Metropolitan area

The digital transportation network data and their road-type hierarchies (highways, boulevards, main streets, streets, and dead-end streets) and the location of medical emergency service stations were obtained from Eskişehir Metropolitan Municipality and integrated into a GIS database to use in GIS-based accessibility network analyses (Figure 5).

The GPS-based floating car data of Eskişehir Metropolitan area were not available in the transportation department of Eskişehir Metropolitan Municipality. Therefore, empirical GPS-based floating car data were collected in Eskişehir city by 1-week fieldwork on August 2007 with sample track intervals of 50 m (Figure 6). A Magellan Explorist 600



Figure 5. The data of digital transportation road network and the medical emergency service locations.

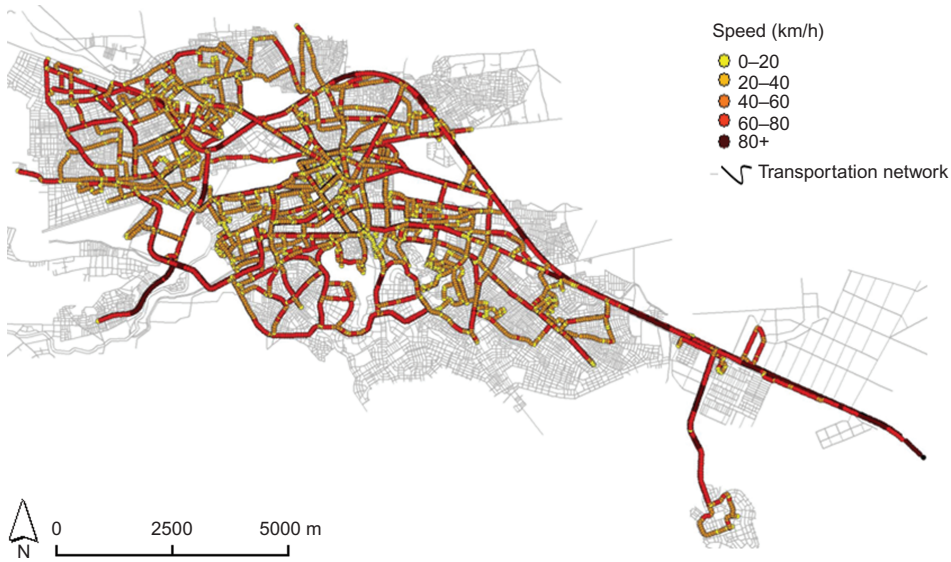


Figure 6. GPS-based floating car traffic data collected by 1 week field work in August 2007.

type GPS receiver was mounted on a probe vehicle and the vehicle location, speed, and time information was regularly recorded between 06:00 a.m. and 23:00 p.m. during 7 days for both peak and normal time periods including weekdays (Monday, Tuesday, Wednesday Thursday, Friday) and weekends (Saturday and Sunday). Most of the main road segments which are highways, boulevards, and main streets are covered in the GPS-based floating car data collection process (nearly 75% of the total). The inner streets and dead-end streets are partly covered (nearly 10% of the total) because of the time and budget limitations of the research. However, the inner streets and dead-end streets do not have a considerable effect on the accessibility results when compared with the highways, boulevards, and main streets. Although a short-term collected empirical GPS-based data are used in the proposed model because of the time, budget, and data availability limitations, it is always possible to implement the model with more detailed and complex traffic data according to aim, budget, and detail needs of the study.

3.2. Implementation of the model

Implementation of the model covers the generation of speed surface from GPS-based floating car data, extraction of speed statistics from the speed surface, and performing MCS. The whole process is implemented by using the ArcGIS Model Builder and MATLAB programming environments.

The average mean and standard deviation statistics extracted from the speed surface for different road types are 53.36 km/h mean speed for highways with 14.51 km/h standard deviation, 43.79 km/h mean speed for boulevards with 14.78 standard deviation, 34 km/h mean speed for main streets with 11.70 standard deviation, 30.96 km/h mean speed for streets with standard deviation 11.79, and 28.96 km/h mean speed for dead-end streets with standard deviation 11.87 (Figure 7).

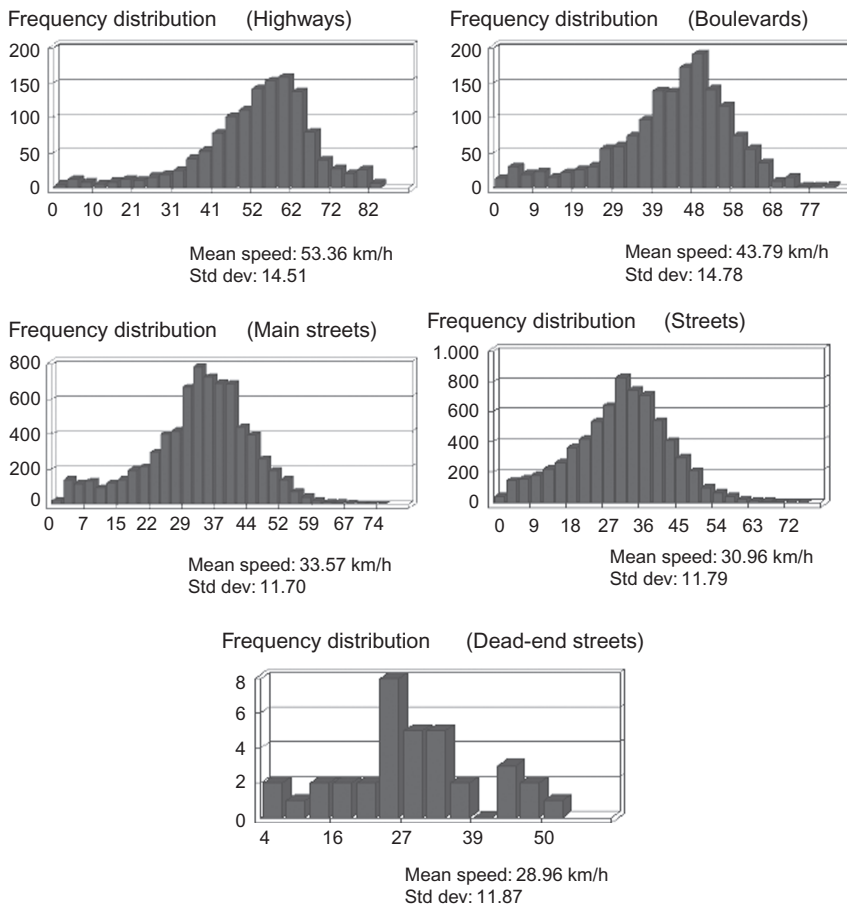


Figure 7. The observed mean and standard deviation statistics for different road types extracted from the GPS-based floating car data.

In the deterministic model, averages of the collected GPS speed data according to the transportation network hierarchies are used which are 53 km/h for highways, 44 km/h for boulevards, 34 km/h for main streets, 31 km/h for streets, and 29 km/h for dead-end streets.

In the stochastic model, random transportation costs produced from mean and standard deviation statistics by MCS are used. During the stochastic modeling process, for the road segments that have GPS data, the local mean and standard deviation statistics are used for generation of random transportation costs. For the road segments that do not have GPS data, the average mean and standard deviation statistics are used for generation of random transportation costs. As the main focus of the study is not to model a specific accessibility condition in a detailed manner, the traffic signal effect on accessibility is not taken into consideration in both of the deterministic and the stochastic model.

The generation of speed surface (with pixel size of 1 m) and extraction of speed statistics from the speed surface took nearly 10 minutes time in Pentium 3 GHz, 3.5 gigabyte RAM desktop computer (there are nearly 14,000 point features in the GPS-based floating car data and 26,000 line features in the transportation network). However, the process time can be less or more depending on the number of features in the GPS-based floating car and

transportation network data, the pixel size of the produced speed surface, and accessibility map and the hardware configuration of the used computer.

3.3. Results

The 5 minutes medical emergency service accessibility is modeled by both of the deterministic model (Figure 8) and the proposed stochastic model (Figure 9). In the stochastic model, a total number of 1000 MCS are performed within nearly 10 hours time and the simulations are cut off after 1000 simulations as a considerable change is not observed in the results (Figure 10).

When the results are overlaid in GIS environment, the differences between deterministic and stochastic models can be observed better. The 5 minutes catchment boundary in stochastic model is significantly different than the deterministic model. Although some of the urban areas in the northwest and south parts of the case study area are observed in the inaccessible zone in the deterministic model, actually they are found in the high-probability accessibility zone (in the 90–100% probability accessibility zone) according to the stochastic model (A zones in Figure 11). Similarly, the northeast and southwest parts of the case study area are observed in the accessible zone in the deterministic model, however they are in the very low probability accessibility zone (in the 20–30% probability accessibility zone) in the stochastic model (B zones in Figure 11). The observed catchment boundary differences can reach up to 2 km length, which can be considered as an important difference from accuracy and reliability point of view and can directly affect/mislead accessibility, location/allocation, and service/catchment area related strategy development and decision-making process.

Finally it can be said that the proposed stochastic model can be considered as more realistic when compared to deterministic model. The stochastic model could handle speed variations in road segments rather than using constant (fixed) deterministic speeds and consider all possible catchment boundaries by the help of the performed simulations. The benefit of the proposed stochastic model is that it could decrease uncertainty in accessibility modeling and increase accuracy and reliability by providing additional probability



Figure 8. Deterministic modeling of accessibility.

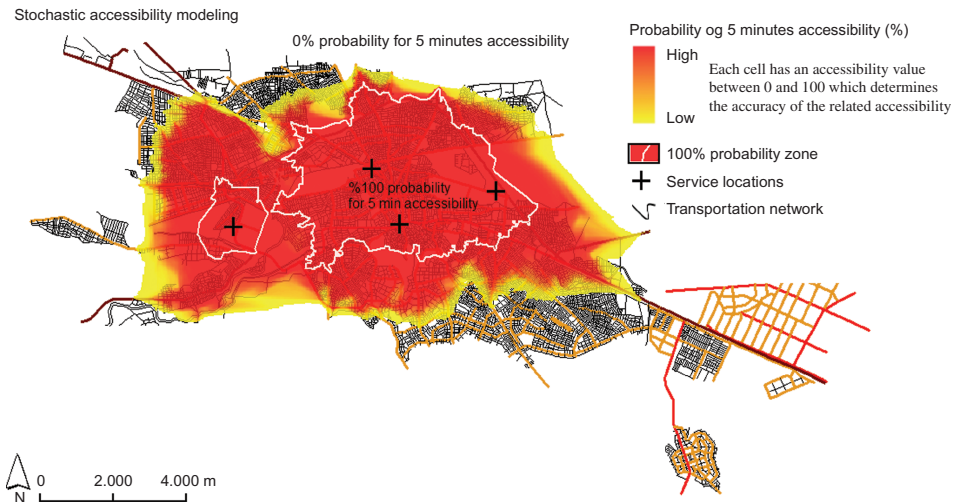


Figure 9. Stochastic modeling of accessibility.

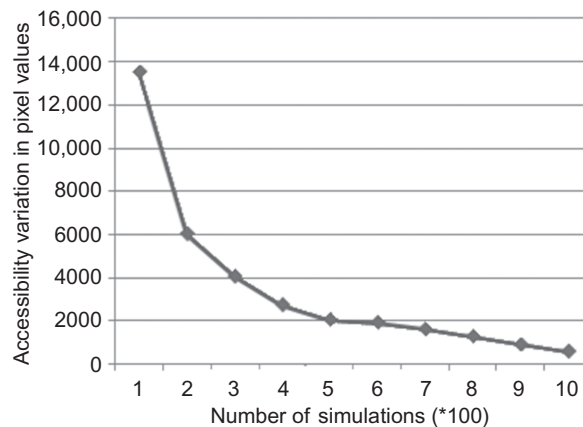


Figure 10. Variation in pixel values with number of simulations.

information related to the catchment boundaries. By this way, it is possible to differentiate regions that have low/moderate/high probability of having 5 minutes of emergency service accessibility or 'there is only a 75% certainty that the place 'x' has 5 minutes emergency service accessibility' can be originally predicted, which is not possible to differentiate in the deterministic model.

4. Conclusions

Compared to previous deterministic methodologies which could only handle constant/fixed transportation costs (e.g., 50 km/h for all main streets) and constant/fixed catchment boundaries (a most likely or average catchment boundary), the proposed stochastic methodology could handle variations in transportation costs and could handle possible catchment

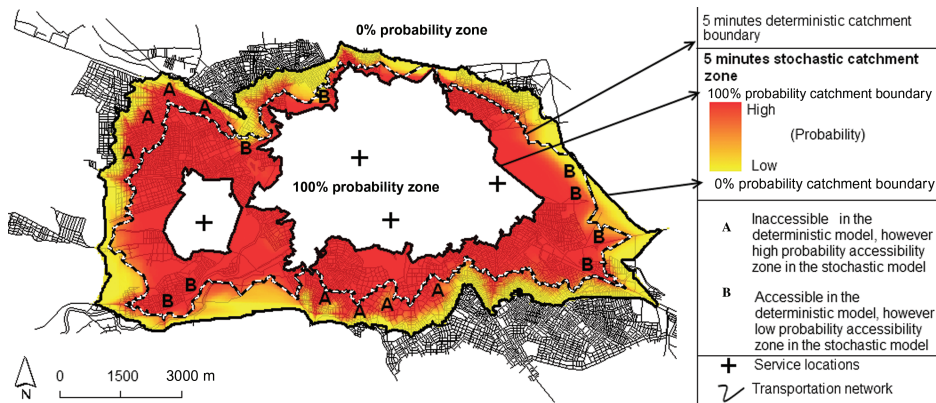


Figure 11. The overlay of stochastic and deterministic modeling of accessibility in GIS environment.

boundaries by the help of the MCS simulations. Hence, the proposed stochastic model provides systematic treatment of uncertainties in the transportation costs, creates a significant improvement on accuracy and reliability of accessibility measures in terms of catchment area boundaries, and provides better decision support for the decision-makers who are supposed to deal with accessibility, location/allocation, and service/catchment area related issues.

Although the proposed stochastic model is performed by using GPS-based floating car data because of its fast and accurate data obtaining and integrating possibilities with GIS, the other traffic data collection methods such as induction loops and optical systems can also be integrated into the stochastic model and provide complementary traffic profile information, where it is not possible or partly possible to obtain satisfactory GPS-based floating car data. After the traffic data, mean, and standard deviation statistics are integrated into a GIS-based digital transportation network, an effective decision support is possible for many accessibility, location/allocation, and service/catchment area related studies.

As the primary focus of this article is not to evaluate a specific accessibility condition in a detailed manner but to provide a methodological discussion and comparison between the deterministic and stochastic accessibility modeling process, the proposed model is implemented by using an empirical short-term collected GPS-based traffic data without considering potential systematic variations (such as rush hour, seasonal, weekend/weekday traffic condition). However, it is very important to stress that traffic can involve potential systematic patterns/variations (such as rush hour, seasonal, day/night, and weekend/weekday). For instance, the rush hour traffic condition can be very different from that at other times. Generalizing whole traffic data in a random process and hiding the potential systematic patterns can be less convincing and informative for the decision-makers who are supposed to deal with accessibility, location/allocation, and service/catchment area related issues. Differentiating the obtained/collected traffic data in terms of potential systematic patterns (such as rush hour, seasonal, day/night, and weekend/weekday) and considering each of the condition separately in a stochastic manner could directly improve the accuracy and reliability of the proposed stochastic model and can be more convincing and informative for the decision-makers who use the model. The detail and complexity of the obtained/collected traffic data and the considered traffic conditions (seasonal, rush hour variations, etc.) can also be rearranged by the decision-makers according to the aim, the budget, and the specific detail needs of their study.

The proposed stochastic methodology could be implemented on modeling of any kind of accessibility measure, ranging from simple travel time measures to more sophisticated cumulative opportunity, gravity, two-step FCA measures, etc. Moreover, the proposed stochastic methodology can easily be adapted to other kinds of accessibility-related studies such as central business district accessibility, job accessibility, recreational accessibility, trade center accessibility, or educational accessibility by considering other several transportation modes such as pedestrian, bicycle, car, or public transport, etc.

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